

physical or life processes. The second law is related to entropy, which is a measure of disorder. An orderly system is associated with entropy minimisation, which means minimisation of energy, space and time for a given amount of effort.

Life is an open system that exchanges energy and information with its surroundings for any effect due to any cause. The second law of thermodynamics confirms that an open or life system can be made more knowledgeable (more orderly or reduced entropy) only by increasing disorder in its surroundings or environment. Thus, knowledge increases order of organisation or, otherwise speaking, minimises organisational consumption of energy, space and time. Therefore justification of Stonier's theory may hold true.

These developments are the manifestation of the basic law of nature, which says, in this universe if anything remains constant, it is nothing but change. Stonier's work on information treats information as the fundamental thing of the universe. And so, information must follow the fundamental laws of the universe, like Newton's laws of motion, mass-energy equivalency, theory of relativity, etc.

Bringing analogy to Newton's laws of motion, it can be said that laws guide information. Before establishing laws, analogies are

defined, as follows:

1. A bit of information is analogically a mass particle. (Here, bit is a bit of information.)
2. Network is the universe of bits.
3. A bit moves with velocity that has the unit of bit/sec, and acceleration with the unit of bit/sec². (Here, bit is a bit of data.)
4. Knowledge is analogous to force. Unit of knowledge is proposed as KN such that,
 $1KN = 1\text{bit} - \text{bit/sec}^2$

First law of information: All bits continue to stay in their files, unless compelled to move by processing by application of knowledge to add value, and vice-versa.

Second law of information: Acceleration (a) of bits for moving in a network derives from application of knowledge (k) by the rule:

$$k = (\text{bits of information}) \times a$$

Third law of information: Every action created by knowledge has an equal and opposite reaction. Computer viruses/security are reactions to the knowledge application on information.

After natural and electrical communication, what is left is mind

communication. Today, one of the main criteria for the development of technological electrical communication is 'fast'.

It is said 'fast' will be last thing for electrical communication to solve. For example, data transfer rates started with a few bits per second, but have now gone up to kilobits (1000) per second to megabits (1,000,000) per second to gigabits (1,000,000,000) to terabits (1,000,000,000,000) per second.

In Navaratnas Sabha, Akbar once asked his Navratanas, "What moves fast?" When eight of nine Ratanas pointed towards the royal horse, Birbal got an edge over the others by saying, 'Our mind.' It has been established that today's fiction becomes tomorrow's reality in science and technology. It will therefore not be unreasonable to state that, in future, technological communication may lead to mind communication, too, which may lead to the Mind Internet. **END** 🐧

This article is dedicated by the author to Sir J.C. Bose, the greatest engineer India has ever produced.

 By: Prof. Chandan Tilak Bhunia

The author is PhD in computer engineering from Jadavpur University. He is also fellow of Institution of Electronics and Telecommunication Engineers, fellow of Institution of Engineers (India) and fellow of Computer Society of India.

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